Makol Chuol

St. Paul, MN | (612) 296-5906 | mdeng1@macalester.edu | linkedin.com/in/makol-jc | github.com/Makoljonchuol

EDUCATION

Macalester College: B.A. Computer Science, Minor: Educational Studies Expected May 2027
GPA: 3.72/4.00 (Spring 2026 GPA: 4.0)
Honors: Kofi Annan Scholar, Dean's List, Fleming First-Generation Scholar, Full-Ride Scholarship
Relevant Coursework: Algorithms, Computer Systems, Data Structures, Discrete Math, CodePath Cybersecurity

SKILLS & CERTIFICATIONS

Cloud & Security: AWS (IAM, ECS, S3, Lambda, CloudWatch, KMS), Terraform, GCP, Azure, OAuth2/OIDC, JWT, SIEM/ELK Stack, NIST Cybersecurity Framework, Wireshark
Languages: Python, Java, C/C++, JavaScript/TypeScript, Bash, Swift
Tools & DevOps: Docker, Kubernetes, Git, REST APIs, Linux (CLI), Keycloak, Temporal, Gunicorn, Locust, Jira
Certifications: CodePath Cybersecurity Certificate (CYB 102), Cisco Cybersecurity Essentials

EXPERIENCE

Software Development Intern | MacroFab | Houston, TX Jun 2026–Aug 2026

- Ranked a top-3 contributor on the engineering team: 1,347 GitHub events, 100+ PRs and 75 Jira issues across authentication, infrastructure, security and performance work
- Rebuilt the PR-review GitHub Action around Pi, a tool-using coding agent (Gemini on Vertex AI), with RAG retrieval via Lambda and Qdrant; adopted team-wide and flagged a production-breaking bug before merge
- Enabled Gunicorn --preload across 20+ Python microservices with post_fork hooks resetting MySQL, SQLAlchemy and boto3 connections; benchmarked each with Locust, raising throughput by up to 22%
- Built OAuth2/JWT auth for Temporal with Keycloak: upgraded to v26.2, wrote Terraform for an HA cluster on ECS Fargate, and added client-credentials auth to the shared Python library
- Replaced long-lived AWS keys with GitHub OIDC federation for Dependabot (Terraform/IAM); built a reusable GitHub Action for Python dead-code detection (Vulture)

Cybersecurity Developer Intern | CrossRealms International | Chicago, IL Feb 2026–May 2026

- Developed scalable telemetry pipelines to detect anomalous activity across cloud and network environments, supporting proactive threat detection and monitoring

Cybersecurity Engineering Intern | CrossRealms International | Chicago, IL May 2025–Aug 2025

- Deployed AI-assisted security monitoring on Azure to detect and classify human, bot and automated attacks
- Automated honeypot deployment across 5 data centers via Terraform and GitHub Actions, cutting setup to < 10 minutes
- Analyzed 278K+ real-world attacks to generate threat intelligence supporting detection and response strategies

Preceptor, Software Design & Development (COMP 225) | Macalester College | St. Paul, MN Aug 2025–Dec 2025

- Mentored 35+ students building secure full-stack applications (React, Node.js, TypeScript, Swift); reviewed code for secure coding practices and maintainability

ACADEMIC PROJECTS

- Personal Portfolio (Astro, TypeScript, GitHub Actions) – makoljonchuol.github.io/makol-chuol-portfolio

LEADERSHIP & ADDITIONAL

- Voices of Amani (Davis Projects for Peace, $10,000 grant)-designed and led a peace education training program at Kakuma Refugee Camp, equipping 12 teachers as "Peace Champions" reaching 300+ students in conflict resolution and cross-cultural understanding
- Macathon 2024-1st Place: led a 5-person team building an energy-generating soccer ball for off-grid refugee schools; pitched on technical feasibility and social impact
- Kakuma Tech Hub–Founding Lead & Curriculum Developer: designed cloud/AI-based digital learning programs and mentored 80+ students on IT fundamentals in a resource-limited setting